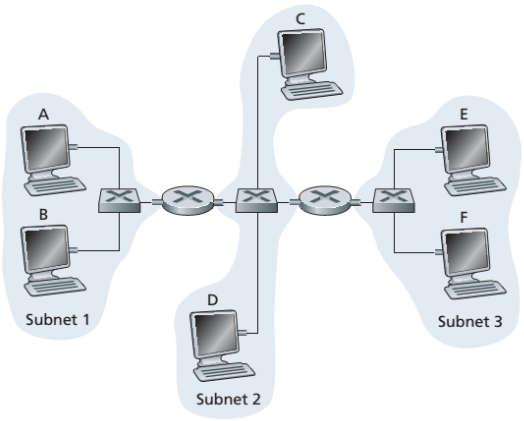
Problem Set: 6
Computer Networks and the Internet: Link Layer

1 Essay Questions

1. What are some of the possible services that a link-layer protocol can offer to the network layer? Which of these link-layer services have corresponding services in IP? In TCP?
2. If all the links in the Internet were to provide reliable delivery service, would the TCP reliable delivery service be redundant? Justify your answer.
3. Why would the token-ring protocol be inefficient if a LAN had a very large perimeter?
4. In Section 5.3, we listed four desirable characteristics of a broadcast channel. Which of these characteristics does slotted ALOHA have? Which of these characteristics does token passing have?

Figure 1: Problem 3



2 Problems

1. Consider the 7-bit generator, G=10011, and suppose that D has the value 1010101010. What is the value of R?
2. A group of N stations shares a 56-kbps pure ALOHA channel. Each station outputs a 1000-bit frame on average once every 100 sec. What is the maximum value of N?

3. In figure 1, suppose that the leftmost router in is replaced by a switch. Hosts A, B, C, and D and the right router are all star-connected into this switch. Give the source and destination MAC addresses, the source and destination IP addresses in the frame encapsulating this IP datagram as the frame is transmitted:
   (a) From A to the switch,
   (b) From the switch to the right router
   (c) From the right router to F.

Good Luck